

Black-Box 5-State a'_x Estimator — Consolidated Current Version

PAUSED (2026-07-06). The production controller stays in the 4-state (no state augmentation) architecture; Q44/R22 are being re-derived directly within it instead of adopting this 5-state augmented-state line as the base — see `reference/eq17_analysis/4state_adet_aprime_noheat_status.md` (a concurrent, parallel-session investigation). This document and `ctrl_const.use_selfmod` (commits 4f5c663, 121f86a) are frozen as a validated reference/comparison point, **not a dead end**: the Sec. 7/10 $Q(a')$ dimensional-anchor + magnitude-gate design, and the κ -sweep methodology exposing the bias-variance / SNR-dwell-time trade-off, are directly reusable if the 4-state line's Q44/R22 re-derivation needs the same technique for its own $\text{Var}(e_{a'})$ term (that document already flags reusing this methodology).

Status (prior, for reference). Consolidates `5state_est_aprime.tex` + `5state_aprime_selfmod.tex` (F_e/H algebra, reused verbatim, Sec. 3–5) with `aprime_blackbox_regression.tex` (Isserlis unbiasedness of a_{xm} , Sec. 2) and pieces from this and a later session: the $Q(a'_x)$ formula (Sec. 7), the δx_r -vs- $\delta \hat{x}$ signal reconciliation (Sec. 6, CONFIRMED as the sign-flip explanation, Sec. 9), a concrete near-wall reversion fix proposal (Sec. 8, still undone), and a self-dither magnitude gate fixing a closed-loop divergence found when the estimator was actually closed into the loop (Sec. 10). **Implemented in code** as `ctrl_const.use_selfmod` in `motion_control_law_eq17_5state_aprime.m` (commit 4f5c663 + the Sec. 10 gate fix); default `false` reproduces the pre-existing baseline bit-for-bit.

1. Setting: the black-box premise

	Known-wall (4state_del_hd_ar1)	4-state	This document (black-box)
a_x predict target	$a_{x_{det}}[k] = a_{nom}/c(\bar{h}_d[k])$		none — integrated state, no external target
a'_x	$-a_x K_h / R$, from the real $c(\bar{h})$		<i>estimated</i> , 5th state
K_h, K'_h	<code>calc_correction_functions</code>		forbidden anywhere, incl. $t=0$ seeding
Spatial memory	—		none; purely time-recursive (no $\bar{h}$ -indexed map)

Wall *location* is known ($\bar{h}[k]$ measurable from $p_m[k]$ and the known wall plane); only the *function* $c(\bar{h})$ is unknown. Excitation requirement (per design discussion): the estimator must work **both** during commanded height motion *and* during a static hold.

2. Why a_{xm} is a legitimate, unbiased measurement (imported result)

$a_{xm}[k] = a_x[k-d] + n_a[k]$ is built by squaring and EWMA-averaging the residual δx_r (`pm_to_axm_derivation.tex`), so a natural worry is circularity: is a_{xm} 's own noise n_a correlated with the very state (a_x , or its fluctuation) we want to measure? `aprime_blackbox_regression.tex` resolved this by Isserlis' theorem (zero-mean jointly Gaussian vectors have zero odd-order moments): every term contributing to a_{xm} is a *quadratic* functional of δx_r , so its covariance with δx_r itself (degree 3, odd) vanishes exactly, regardless of a_{cov} , the LP bandwidth, or δx_r 's autocorrelation structure. This is why a_{xm} can be treated below as a clean linear measurement of $a_x[k-d]$ with additive noise n_a , not a self-referential trap.

3. Parameter model (imported from 5state_aprime_selfmod, verbatim)

Local decomposition (derivation device only — **not** evaluated at runtime):

$$a_x[k] = a_{\text{det}}[k] + a_{x_{ram}}[k], \quad a_{x_{ram}}[k] = a'_x x_{ram}[k] = -a'_x \delta x[k]$$

Closing the loop ($\delta x_3 = \delta x[k]$ the current tracking error, ε the lumped closed-loop noise defined in Sec. 4):

$$\delta x[k+1] = \lambda_c \delta x_3[k] - \varepsilon[k] \Rightarrow \Delta \delta x[k] = -(1 - \lambda_c) \delta x_3[k] - \varepsilon[k]$$

$$\begin{cases} a_x[k+1] = a_x[k] + a'_x [\Delta h_d[k] + (1 - \lambda_c) \delta x_3[k]] + w_4[k] \\ a'_x[k+1] = a'_x[k] + w_{a'}[k] \end{cases}$$

$\Delta h_d[k] = h_d[k+1] - h_d[k]$ (known, from the commanded trajectory) supplies observability **during motion**; $(1 - \lambda_c) \delta x_3[k]$ (thermal self-dither, zero-mean, $\sim$ tens of nm) supplies observability **on a hold**. Neither requires $c(\bar{h})$.

4. State equation (5-state)

$$\delta x_1[k+1] = \delta x_2[k], \quad \delta x_2[k+1] = \delta x_3[k],$$

$$\delta x_3[k+1] = \lambda_c \delta x_3[k] - F_{dx}[k] e_{a_x}[k] + dF_{dx}^h[k] e_{a'}[k] - \varepsilon[k],$$

$$a_x[k+1] = a_x[k] + a'_x [\Delta h_d[k] + (1 - \lambda_c) \delta x_3[k]] + w_4[k],$$

$$a'_x[k+1] = a'_x[k] + w_{a'}[k].$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i], \quad dF_{dx}^h[k] = (1 - \lambda_c) \sum_{i=1}^d (h_d[k] - h_d[k-i]) f_{dx}[k-i]$$

$$e_{a_x} = a_x - \hat{a}_x, \quad e_{a'} = a'_x - \hat{a}'_x \text{ (level and slope estimation errors).}$$

5. Output equation, $\mathbf{H}$, closed-loop estimator, $\mathbf{F}_e$ (imported verbatim)

$$a_x[k-d] = a_x[k] - a'_x [\Delta H_d[k] - (\delta x_3[k] - \delta x_1[k])], \quad \Delta H_d[k] = h_d[k] - h_d[k-d]$$

$$y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]},$$

$$y_2[k] = a_{xm}[k] = a_x[k-d] + n_a[k] \approx a_x[k] - a'_x [\Delta H_d[k] - (\delta x_3[k] - \delta x_1[k])] + \underbrace{n_a[k]}_{r_2[k]}.$$

The span $a'_x(\delta x_3 - \delta x_1)$ is carried entirely by $\mathbf{H}$ (not left in r_2) — this is the fix for the "common thermal noise double-count" that self-mod's Caveat 2 documents (keeping the span in *both* $\mathbf{H}$ and R_{22} self-defeats the very observability the self-dither term is supposed to provide).

$$\mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -\hat{a}'_x & 0 & \hat{a}'_x & 1 & -\Delta H_d[k] + (\delta \hat{x}_3 - \delta \hat{x}_1) \end{bmatrix}$$

($\mathbf{H}(2, 1), \mathbf{H}(2, 3) = \mp \hat{a}'_x$ are $O(\hat{a}'_x)$, negligible; the load-bearing term is $\mathbf{H}(2, 5)$.)

Error dynamics ($G_4[k] := \Delta h_d[k] + (1 - \lambda_c)\delta \hat{x}_3[k]$, $G_H[k] := \Delta H_d[k] - (\delta \hat{x}_3 - \delta \hat{x}_1)$):

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & dF_{dx}^h[k] \\ 0 & 0 & 0 & 1 & G_4[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$\mathbf{F}_e(4, 5) = G_4[k]$ is the direct consequence of a_x 's own predict equation using $\hat{a}'_x$ (Sec. 3) — *not* an assumption bolted on. $\mathbf{F}_e(3, 4) = -F_{dx}$ is unchanged from the 4-state (control law uses $\hat{a}_x$ to compute force). a'_x affects δx_3 only *indirectly*, through a_x 's error (two-hop propagation); there is no direct $\mathbf{F}_e(3, 5)$ unless $\hat{a}'_x$ is also fed forward into the control law directly (a separate design choice, not made here).

6. Reconciliation: raw δx_r vs. filtered $\delta \hat{x}_3 - \delta \hat{x}_1$ — leading explanation for the sign-flip

This session’s external, standalone regression test (`verify_aprime_blackbox_v1_ensemble.m`) computed

$$\hat{a}'_{\text{ext}} = -\frac{\text{Cov}(a_{xm}, \delta x_r)}{\text{Var}(\delta x_r)}$$

using the **raw** residual δx_r (the a_{xm} channel’s own IIR intermediate, *before* any state filtering) as the Δh proxy. Against 60 independent seeds (static hold, $\bar{h} \approx 1.78$), this gave a statistically significant **wrong-sign** result (t -stat -2.9 to -7.6), even though: (i) the pure regression algebra was verified unbiased on synthetic data built to the same model (`verify_aprime_blackbox_synthetic.m`, both i.i.d. and autocorrelated δx_r), (ii) a d -step delay misalignment does not by itself flip the sign (`verify_aprime_blackbox_synthetic_delay.m`), and (iii) an oracle indexing fix (pre/post-integration alignment) changed nothing.

The present derivation (Sec. 5, imported from `5state_aprime_selfmod`) uses $\delta \hat{x}_3 - \delta \hat{x}_1$ — the **EKF’s own filtered state estimates** of $\delta x_3, \delta x_1$ — not the raw δx_r . These are related but **not** the same signal: $\delta \hat{x}_3, \delta \hat{x}_1$ are shaped by the *closed-loop feedback* (the control law computes $f_{dx}[k]$ directly from $\delta x_m[k]$, which the raw δx_r also feeds into, but $\delta \hat{x}$ is the KF’s own posterior after fusing y_1 and y_2 , weighted by the current **P**). **Leading (unconfirmed) hypothesis:** the external test’s raw- δx_r regressor absorbs contamination from this closed-loop feedback path (the control law’s use of $\delta x_m[k]$ to compute the very force that determines the next position) that the properly-derived **H**(2, 5) — built from the filtered $\delta \hat{x}$, and appearing *inside* a jointly-consistent **F_e/H** pair — does not suffer, because the KF’s own error-dynamics bookkeeping (Sec. 5) already accounts for this coupling structurally rather than assuming it away via an external statistical identity.

Confirmed (`verify_5state_aprime_selfmod_signflip.m`, same static-hold scenario, real controller with `use_selfmod=true`): at 60 seeds, $\hat{a}'_x$ (read directly from `diag.delta_a_hat`) already showed $|t\text{-stat}| < 1.51$ on all three axes versus the external test’s -2.9 to -7.6 ; the y, z axes matched the oracle’s sign, while x ’s point estimate was nominally opposite but statistically indistinguishable from zero (SE exceeded $|\text{mean}|$). A re-run at 200 seeds resolved this: all three axes now match the oracle’s sign (x, y, z : $t = -0.62, 0.01, -1.24$), confirming the x -axis discrepancy was sampling noise, not a residual sign problem. This confirms the raw-vs-filtered-signal distinction (this section’s hypothesis) was the dominant explanation for the external test’s wrong sign.

7. $Q(a'_x) = q_5$ — honest process-noise design (NEW this session)

`5state_aprime_selfmod.tex` leaves $q_5 = \text{Var}(w_{a'})$ labelled only as “the estimator bandwidth on the slope,” with no formula. Decompose:

$$q_5[k] = \sigma_{a''}^2 \cdot (\Delta h_d[k]^2 + \sigma_{\delta h}^2[k]), \quad \sigma_{\delta h}^2[k] = 4k_B T \hat{a}_x[k]$$

Rationale. a'_x changes because the (unknown) curvature $a''_x = da'_x/d\bar{h}$ is nonzero; moving by Δh changes a'_x by $\approx a''_x \cdot \Delta h$ (chain rule), so $\text{Var}(\Delta a'_x) \approx a''_x{}^2 \cdot \text{Var}(\Delta h)$. $\Delta h_d[k]$ (commanded) is exactly known; $\sigma_{\delta h}^2[k]$ (thermal) reuses the already-validated $4k_B T a_x$ formula with the **current level estimate** $\hat{a}_x[k]$ (honest, no wall peek). Only $\sigma_{a''}^2$ — one scalar, the curvature magnitude prior — remains free.

Dimensional anchor for $\sigma_{a''}^2$

R (probe radius) and $a_{nom} = \Delta t / \gamma_N$ are known probe/fluid constants, **not** wall properties. Since $c(\bar{h})$ is defined as a function of $\bar{h} = h/R$, R is the wall effect’s own natural length scale:

$$\sigma_{a'} \sim a_{nom}/R \quad (\text{slope scale}), \quad \sigma_{a''} \sim \sigma_{a'}/R = a_{nom}/R^2 \quad (\text{curvature scale})$$

$$\sigma_{a''}^2 \sim (a_{nom}/R^2)^2$$

This is an order-of-magnitude engineering prior, not an exact derivation — but it is *derived from known constants*, not picked from nothing.

Empirical check (verify_aprime_Qprime_assumptions.m, run before this file was written)

	near-wall ($\bar{h} \approx 1.05\text{--}1.2$)	$\bar{h} \approx 1.78$	far-field ($\bar{h} > 5$)
$ a'_{true} /(a_{nom}/R)$	0.6–0.9	0.30	≤ 0.04
$ a''_{true} /(a_{nom}/R^2)$	2.2–3.8	0.31	≤ 0.02
$\text{Var}(\Delta a'_{true})$ vs. predicted (real osc trajectory)	ratio 0.65	—	ratio $\sim 10^{-4}$

Near-wall (where a'_x matters most): the anchor is the correct order of magnitude (ratios $0.6\text{--}3.8\times$, and the $\text{Var}(\Delta a')$ check lands within 35%). **Far-field: the fixed constant over-covers by $\sim 10^4\times$** — the already-known, architecturally unavoidable single-constant trade-off (Sec. 8 of `5state_aprime_selfmod.tex`’s antecedent discussion), now quantified with real numbers. Over-covering far-field is low-cost: $a'_x \rightarrow 0$ there and tracking does not depend on it.

8. Caveat 5 fix proposal: state-computable near-wall reversion (NEW this session)

5state_aprime_selfmod.tex Caveat 5: $\mathbf{F}_e(4,4) = 1$ is a free integrator (no reversion target, since there is no known a_{det}); when the near-wall gate turns a_{xm} off, a_x has neither reversion nor measurement and drifts as an undamped random walk. It explicitly flags the fix as required-but-undone: “*a state-computable reversion ($a_{xram} = -a'_x \delta x_3$) ... is required before closing the loop.*”

Proposal

Use the **already-known local-linear relationship** (Sec. 3) as a self-consistency check, not an external target. Define a slow reference level $\bar{a}_x[k]$ (its own, slower EWMA of $\hat{a}_x$ — purely internal, no wall model), and add a weak correction pulling $\hat{a}_x$ toward the value **implied by the current state itself**:

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \hat{a}'_x[k] [\Delta h_d[k] + (1 - \lambda_c) \delta \hat{x}_3[k]] + \kappa \left(\underbrace{\bar{a}_x[k] - \hat{a}'_x[k] \delta \hat{x}_3[k]}_{\text{state-implied } a_{\text{det}} \text{ estimate}} - \hat{a}_x[k] \right)$$

κ small (weak pull, e.g. $\kappa \sim (1 - \lambda_c)/N$ for some large averaging horizon N — a design choice, not yet tuned). Physical bound as a secondary safety net (fully honest, R/a_{nom} are probe constants): $0 < a_x \leq a_{nom}$ always (the wall only adds drag, never removes it below the free-space value); a clip or soft-barrier reparametrization on $\hat{a}_x$ prevents runaway growth even if κ is mistuned. **Not yet derived to the same rigor as Sec. 3–5; flagged here as the next piece of derivation work, not a finished result.**

9. Consolidated verification status

Test	What it checked	Result
V0 (free-space null)	Isserlis unbiasedness on real simulated signal	PASS
V1 (1 Hz osc, single traj.)	external regression vs. oracle	bug found (wrong regressor axis) + locality-vs-SNR tension found
V1-hold (static, single traj.)	same, locality removed	weak corr., large bias; ambiguous (noise vs. real)
V1-ensemble (60 seeds, batch)	same, high power	significant wrong-sign ($t = -2.9$ to -7.6); resolved by the two rows below
Synthetic (pure algebra)	regression formula + Isserlis, $\rho=0, 0.85$	PASS (rules out formula bug)
Synthetic+delay	d -step misalignment alone	does not flip sign (rules out this cause alone)
Oracle pre/post-integration fix	indexing bug	no numerical effect (rules out this cause)
$Q(a')$ (Sec. 7)	magnitude dimensional anchor vs. oracle	near-wall PASS (order of magnitude); far-field over-covers (expected)
5-state (60 seeds, real controller)	<code>use_selfmod</code> same scenario, real <code>diag.delta_a_hat</code>	sign correct on y, z ; x nominally wrong but $ t < 1$ (noise, $SE > mean $)
5-state (200 seeds, real controller)	<code>use_selfmod</code> same scenario, higher power	sign CONFIRMED correct on all 3 axes ($t = -0.62, 0.01, -1.24$)

Resolved: the wrong-sign result was caused by the raw- δx_r -vs-filtered-signal distinction (Sec. 6). The properly-derived 5-state EKF (Sec. 3–5,7), read directly from the real controller (`use_selfmod=true`), recovers the correct sign on all three axes at 200 seeds. **Not yet done:** accuracy characterization beyond sign (L1-style κ sweep, hold vs. motion, near vs. far field); the Sec. 8 near-wall reversion proposal remains undone (this test’s scenario does not exercise the gate-off condition); Q33/Q44/R22 remain wall-peeking (Sec. 7 note).

10. L2 closed-loop divergence: Q(5,5) far-field over-coverage, fixed with a self-dither gate

With `use_selfmod=true` closing the loop for real (`use_true_gain=false`: the control law uses $\hat{a}_x$, not the true gain), a 1 Hz near-wall oscillation scenario (`verify_eq17_5state_aprime_selfmod_L1.m`) showed the open-loop estimability gain (Sec. 9-adjacent open-loop test: $\Delta\text{corr}_z \approx +0.79$, zero tracking cost) does **not** carry over unmodified: z -axis tracking std nearly doubled ($26.9 \rightarrow 52.3$ nm) and 1 of 10 seeds diverged, while the same toggle left off never diverged.

Root cause

The divergence occurred not near the wall but during the trajectory’s fast DESCENT transient (large commanded $|\Delta h_d|$, far from the wall, $\bar{h} \approx 6\text{--}15$), traced to $\hat{a}_x$ drifting monotonically off its true value over dozens of steps. A magnitude gate on $\mathbf{F}_e(4, 5)/\mathbf{H}(2, 5)$ alone (closing them whenever $|\Delta h_d|$ dominates the thermal scale) did **not** fix it — tracing individual steps showed the gate was already correctly closed throughout the bad region, ruling out the self-dither coupling terms as the cause. The actual culprit is q_5 (Sec. 7): the honest anchor $\sigma_{a''}^2 \sim (a_{nom}/R^2)^2$ has **no wall-distance awareness**, unlike the baseline $a'' = (a/R^2)(K_h^2 - K_h')$ it replaces (which shrinks far from the wall via $K_h, K_h' \rightarrow 0$). During fast motion far from the wall, q_5 ’s Δh_d^2 term over-inflates by $\sim 10^4\text{--}10^5\times$ — exactly the already-documented far-field over-coverage (Sec. 7’s empirical table), previously judged “low-cost” *under open-loop scoring, where the over-covered $\hat{a}'_x$ never touches real dynamics*. In closed loop, $\hat{a}'_x$ ’s resulting noise multiplies $\Delta h_d[k]$ directly in $\hat{a}_x[k+1] = \hat{a}_x[k] + \hat{a}'_x[k]\Delta h_d[k]$, corrupting $\hat{a}_x$ and hence $f_d = 1/\hat{a}_x\{\dots\}$ itself.

Fix

Extend the same honest, no- $c(\bar{h})$ magnitude gate to **both** the self-dither coupling terms AND q_5 ’s Δh_d^2 term:

$$\boxed{\begin{aligned} \text{gate}_i[k] &= (|\Delta h_d[k]| < \sigma_{\delta h, i}[k]) \\ \sigma_{\delta h, i}[k] &= \sqrt{4k_B T \hat{a}_{x, i}[k]} \end{aligned}}$$

Gate open (thermal-dominated, the regime self-dither was designed for): $\mathbf{F}_e(4, 5)$, $\mathbf{H}(2, 5)$, and q_5 all use their full self-mod forms (Sec. 5, Sec. 7). Gate closed (motion-dominated, e.g. a fast descent): $\mathbf{F}_e(4, 5) = \Delta h_d$, $\mathbf{H}(2, 5) = -\Delta H_d$ (reduce to baseline), and $q_5 = \sigma_{a''}^2 \cdot \sigma_{\delta h}^2$ (drop the Δh_d^2 term only — still the honest anchor, not a wall-peeking fallback).

Verification

Same 1 Hz scenario, 10 seeds, closed loop: **no divergence** (was 1 of 10 seeds); $\Delta(\text{track_std}_z) = +4.1$ nm (was +25 to +33 nm), a modest cost consistent with the already-known near-wall spread increase; $\Delta\text{corr}_z = +0.74$ preserved. The static-hold sign-confirmation scenario (Sec. 6/9, 60 and 200 seeds) re-run **bit-for-bit identical** to before this fix, as expected: $\Delta h_d \approx 0$ throughout a hold keeps the gate open, so nothing there changes. `verify_eq17_5state_aprime_selfmod_L0.m` Check 3 added to lock in the gate’s open/closed logic against regression.

Lesson: an “honest, no wall-peek” formula can still be wrong in a way pure open-loop testing cannot reveal — the far-field over-coverage was known and judged low-cost *under an evaluation regime that*

never let the over-covered quantity touch real dynamics. Closing the loop for real (L2) is not optional follow-on characterization; it is precisely where this class of bug lives, and it should be run before any claim that an estimator improvement helps tracking.

Not yet done: L2 testing so far covers only this one trajectory (1 Hz near-wall oscillation with a fast pre-oscillation descent); other fast-transient shapes are untested. Whether feeding $\hat{a}'_x$ -improved $\hat{a}_x$ into the control law yields a measurable tracking benefit anywhere (vs. merely “no regression”) remains open.